

Week 1 Module

- Programming Language - Java

Star patterns :->

Because we are making 2-D star pattern that's why we need minimum two loops (eg. while, for).

I am writing code for dynamic pattern printing so in my program first I take input from user and then according to the user's input my code generates the pattern.

- Let's know about our variables -

$K \rightarrow$ K variable stores the input

$i \rightarrow$ i is parent loop variable

$j \rightarrow$ j is child loop variable.

Code & Output :-

Question- 01

```
Pattern > left_up.java
1  import java.util.Scanner;
2
3  public class left_up{
4      public static void main (String[] args){
5
6          System.out.println("Enter Grid Size");
7          Scanner sc = new Scanner (System.in);
8          int k = sc.nextInt();
9          System.out.println("Here is your pattern");
10
11         for (int i = 0 ; i < k ; i++){
12
13             for( int j = 0 ; j <= i ; j++ ){
14                 System.out.print("*");
15             }
16
17             System.out.println(" "); // line chang krne ke liye
18         }
19     }
20 }
21
```

PROBLEMS OUTPUT DEBUG CONSOLE **TERMINAL** Code + -

```
PS C:\Users\Weeraj\Desktop\java> cd "c:\Users\Weeraj\Desktop\java\Pattern\" ; if ($?) { javac left_up.java } ; if ($?) { java left_up }
Enter Grid Size
5
Here is your pattern
*
**
***
****
*****
PS C:\Users\Weeraj\Desktop\java\Pattern>
```

Question- 02

```
Pattern > squire.java
1  import java.util.Scanner;
2
3  public class squire{
4      public static void main(String[] args){
5
6          System.out.println("Enter Grid Size");
7          Scanner sc = new Scanner (System.in);
8          int k = sc.nextInt();
9          System.out.println("Here is your pattern");
10
11         for ( int i = 0 ; i < k ; i++){
12
13             for( int j = 0 ; j < k ; j++){
14                 System.out.print("*");
15             }
16             System.out.println("");
17         }
18     }
19 }

```

PROBLEMS OUTPUT DEBUG CONSOLE **TERMINAL** Code + -

```
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PS C:\Users\Weeraj\Desktop\java> cd "c:\Users\Weeraj\Desktop\java\Pattern\" ; if ($?) { javac squire.java } ; if ($?) { java squire }
Enter Grid Size
5
Here is your pattern
*****
*****
*****
*****
*****
PS C:\Users\Weeraj\Desktop\java\Pattern>
```

Question -03

```

Pattern > box_bordarjava
1  import java.util.Scanner;
2
3  public class box_bordar{
4      public static void main (String[] args){
5
6          System.out.println("Enter the Grid Size");
7          Scanner sc = new Scanner(System.in);
8          int k = sc.nextInt();
9          System.out.println("Here is your pattern");
10
11         for (int i = 0 ; i < k ; i++){
12
13             for(int j = 0 ; j < k ; j++){
14                 if (j == 0 || j == k-1){
15                     System.out.print("*");
16                 }
17                 else{
18                     if(i == 0 || i == k-1){
19                         System.out.print("*");
20                     }
21                     else{
22                         System.out.print(" ");
23                     }
24                 }
25             }
26             System.out.println("");
27         }
28     }
29 }
30

```

```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL

PS C:\Users\Weeraj\Desktop\java\Pattern> cd "c:\Users\We
r }
Enter the Grid Size
5
Here is your pattern
*****
* *
* *
* *
*****
PS C:\Users\Weeraj\Desktop\java\Pattern>

```

Question -04

```

Pattern > left_down.java
1  import java.util.Scanner;
2
3  public class left_down{
4      public static void main (String[] args){
5
6          System.out.println("Enter Grid Size");
7          Scanner sc = new Scanner (System.in);
8          int k = sc.nextInt();
9          System.out.println("Here is your pattern");
10
11         for (int i = 0 ; i < k ; i++){
12
13             for (int j = k ; j > i ; j--){
14                 System.out.print("*");
15             }
16
17             System.out.println(" "); // line chang krne ke liye
18         }
19     }
20 }
21
22

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```

*****
PS C:\Users\Weeraj\Desktop\java\Pattern> cd "c:\Users\Weeraj\Desktop\java\Pattern\" ; if ($?) { javac left_down.java }
Enter Grid Size
5
Here is your pattern
*****
****
***
**
*
PS C:\Users\Weeraj\Desktop\java\Pattern>

```

Question -05

```

Pattern > pyramid.java
1  import java.util.Scanner;
2
3  public class pyramid{
4      public static void main (String[] args){
5
6          System.out.println("Enter Grid Size");
7          Scanner sc = new Scanner (System.in);
8          int k = sc.nextInt();
9          System.out.println("Here is your pattern");
10
11         for(int i = 0 ; i < k ; i++){
12
13             for(int j = k - i; j > 0; j--){
14                 System.out.print(" ");
15             }
16
17             for(int j = 0; j < ((i*2) + 1); j++){
18                 System.out.print("*");
19             }
20
21
22             System.out.println("");
23
24         }
25
26     }
27 }

```

```

PS C:\Users\Weeraj\Desktop\java\Pattern> cd "c:\Users\We
Enter Grid Size
5
Here is your pattern
    *
   ***
  *****
 *****
*****
PS C:\Users\Weeraj\Desktop\java\Pattern>

```

Question -06

```

Pattern > Diamond.java
1  import java.util.Scanner;
2
3  public class Diamond{
4      public static void main (String[] args){
5
6          System.out.println("Enter Grid Size");
7          Scanner sc = new Scanner (System.in);
8          int k = sc.nextInt();
9          System.out.println("Here is your pattern");
10
11         for(int i = 0 ; i < k ; i++){ // pyramid
12
13             for(int j = k - i; j > 0; j--){
14                 System.out.print(" ");
15             }
16
17             for(int j = 0; j < ((i*2) + 1); j++){
18                 System.out.print("*");
19             }
20
21             System.out.println("");
22         }
23
24         for(int i = k ; i >= 0 ; i--){ // up-side down pyramid
25
26             for(int j = k - i; j > 0; j--){
27                 System.out.print(" ");
28             }
29
30             for(int j = 0; j < ((i*2) + 1); j++){
31                 System.out.print("*");
32             }
33
34             System.out.println("");
35         }
36
37     }
38
39 }
40
41 }

```

```

PS C:\Users\Weeraj\Desktop\java\Pattern> cd "c:\Use
Enter Grid Size
5
Here is your pattern
    *
   ***
  *****
 *****
*****
*****
*****
*****
    *
   ***
    *
PS C:\Users\Weeraj\Desktop\java\Pattern>

```


Question -07

```
Pattern > half_diamond.java
1  import java.util.Scanner;
2
3  public class half_diamond{
4      public static void main (String[] args){
5
6          System.out.println("Enter Grid Size");
7          Scanner sc = new Scanner (System.in);
8          int k = sc.nextInt();
9          System.out.println("Here is your pattern");
10
11         for (int i = 0 ; i < k ; i++){ // left up pattern
12             for( int j = 0 ; j <= i ; j++ ){
13                 System.out.print("*");
14             }
15             System.out.println(" "); // line chang krne ke liye
16         }
17         for (int m = 0 ; m < k ; m++){ // left down pattern
18             for (int n = k ; n > m+1 ; n--){
19                 System.out.print("*");
20             }
21             System.out.println(" "); // line chang krne ke liye
22         }
23     }
24 }
25
26
27
28 }
```

```
*
PS C:\Users\Weeraj\Desktop\java\Pattern> cd "c:\Us
amond }
Enter Grid Size
5
Here is your pattern
*
**
***
****
*****
****
***
**
*
PS C:\Users\Weeraj\Desktop\java\Pattern>
```

Number Pattern :->

I am going to use same approach which is i used in Star pattern

But...

here we have to print numbers insed of stars so.

Problem	Approach
• Square	Print "1" in place of Star
• Incrementing Right triangle	print j variable
• Inverted Diamond Sandwich Pattern Incrementing Half Diamond	new variable L ⇒ 3 starting value.

Question -01

```

Pattern > number_box.java
1  import java.util.Scanner;
2
3  public class number_box{
4      public static void main(String[] args){
5
6          System.out.println("Enter Grid Size");
7          Scanner sc = new Scanner (System.in);
8          int k = sc.nextInt();
9          System.out.println("Here is your pattern");
10
11         for (int i = 0 ; i < k ; i++){
12
13             for (int j = 0 ; j < k ; j++){
14                 System.out.print("1"); // in that we have to print "1"
15             }
16             System.out.println(" ");
17         }
18
19     }
20 }
21

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

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```

PS C:\Users\Weeraj\Desktop\java> cd "c:\Users\Weeraj\Desktop\java\Pattern\" ; if ($?) { javac number_box.java }
Enter Grid Size
5
Here is your pattern
11111
11111
11111
11111
11111
11111
PS C:\Users\Weeraj\Desktop\java\Pattern>

```

Question -02

```
Pattern > incrementing_num.java
1  import java.util.Scanner;
2
3  public class incrementing_num{
4      public static void main(String[] args){
5
6          System.out.println("Enter Grid Size");
7          Scanner sc = new Scanner (System.in);
8          int k = sc.nextInt();
9          System.out.println("Here is your pattern");
10
11         for(int i = 0 ; i < k ; i++){
12
13             for(int j = 0 ; j < k ; j++){
14                 System.out.print(j+1);
15             }
16             System.out.println("");
17         }
18     }
19 }
20 }
21 // output for grid size 5
```

PROBLEMS OUTPUT DEBUG CONSOLE **TERMINAL**

```
11111
PS C:\Users\Weeraj\Desktop\java\Pattern> cd "c:\Users\Weeraj\Desktop\java\Pattern\" ; if ($?) { javac incrementing_num.java }
rementing_num }
Enter Grid Size
5
Here is your pattern
12345
12345
12345
12345
12345
PS C:\Users\Weeraj\Desktop\java\Pattern>
```

Question -03

```
Pattern > left_rightangle_triangle.java
1  import java.util.Scanner;
2
3  public class left_rightangle_triangle{
4      public static void main(String[] args){
5
6          System.out.println("Enter Grid Size");
7          Scanner sc = new Scanner (System.in);
8          int k = sc.nextInt();
9          System.out.println("Here is your pattern");
10
11         for(int i = 0 ; i < k ; i++){
12
13             for(int j = 0 ; j <= i ; j++){
14                 System.out.print(j+1);
15             }
16             System.out.println("");
17         }
18     }
19 }
20 }
21
```

PROBLEMS OUTPUT DEBUG CONSOLE **TERMINAL**

```
12345
PS C:\Users\Weeraj\Desktop\java\Pattern> cd "c:\Users\Weeraj\Desktop\java\Pattern\" ; if ($?) { javac left_rightangle_triangle.java }
java left_rightangle_triangle }
Enter Grid Size
5
Here is your pattern
1
12
123
1234
12345
PS C:\Users\Weeraj\Desktop\java\Pattern>
```

Question -04

```

Pattern > down_rightangle_triangle.java
1  import java.util.Scanner;
2
3  public class down_rightangle_triangle{
4      public static void main(String[] args){
5
6          System.out.println("Enter Grid Size");
7          Scanner sc = new Scanner (System.in);
8          int k = sc.nextInt();
9          System.out.println("Here is your pattern");
10
11         int l = 2;
12         for(int i = 0 ; i < k ; i++){
13             l = l+1;
14             for(int j = k ; j > i ; j--){
15
16                 System.out.print(l);
17             }
18             System.out.println("");
19         }
20
21     }
22 }

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```

PS C:\Users\Weeraj\Desktop\java\Pattern> cd "c:\Users\Weeraj\Desktop\java\Pattern\" ; if ($?) { javac down_rightangle_triangle.java
java down_rightangle_triangle }
Enter Grid Size
5
Here is your pattern
33333
4444
555
66
7
PS C:\Users\Weeraj\Desktop\java\Pattern>

```

Question -05

```

Pattern > sandwich.java
1  import java.util.Scanner;
2
3  public class sandwich{
4      public static void main (String[] args){
5
6          System.out.println("Enter Grid Size");
7          Scanner sc = new Scanner (System.in);
8          int k = sc.nextInt();
9          System.out.println("Here is your pattern");
10
11         int l = 2;
12
13         for(int i = k ; i >= 0 ; i--){
14             l = l+1;
15             for(int j = k - i; j > 0; j--){
16                 System.out.print(" ");
17             }
18
19             for(int j = 0; j < ((i*2) + 1); j++){
20                 System.out.print(l);
21             }
22
23             System.out.println("");
24
25             for(int i = 1 ; i < k+1 ; i++){
26                 l = l-1;
27                 for(int j = k - i; j > 0; j--){
28                     System.out.print(" ");
29                 }
30
31                 for(int j = 0; j < ((i*2) + 1); j++){
32                     System.out.print(l);
33                 }
34
35                 System.out.println("");
36
37             }
38         }
39     }
40
41 }
42

```

```

/
PS C:\Users\Weeraj\Desktop\java\Pattern> cd "c:
Enter Grid Size
2
Here is your pattern
33333
444
5
444
33333
PS C:\Users\Weeraj\Desktop\java\Pattern>

```


Question -06

```
Pattern > incrementing_diamond.java
1  import java.util.Scanner;
2
3  public class incrementing_diamond{
4      public static void main(String[] args){
5
6          System.out.println("Enter Grid Size");
7          Scanner sc = new Scanner (System.in);
8          int k = sc.nextInt();
9          System.out.println("Here is your pattern");
10
11         int l = 3;
12         for (int i = 0 ; i < k ; i++){ // left up pattern
13             l = 3+i;
14             for( int j = 0 ; j <= i ; j++){
15
16                 System.out.print(l);
17
18             }
19             System.out.println(" "); // line chang krne ke liye
20         }
21         for (int m = 0 ; m < k ; m++){ // left down pattern
22
23             l = l-1;
24
25             for (int n = k ; n > m+1 ; n--){
26                 System.out.print(l);
27             }
28
29             System.out.println(" "); // line chang krne ke liye
30         }
31     }
32 }
33 }
```

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```
38 // 555
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

incrementing_diamond }
Enter Grid Size
5
Here is your pattern
3
44
555
6666
7777
6666
555
44
3
```